

Station selection

one of five systems, one station each

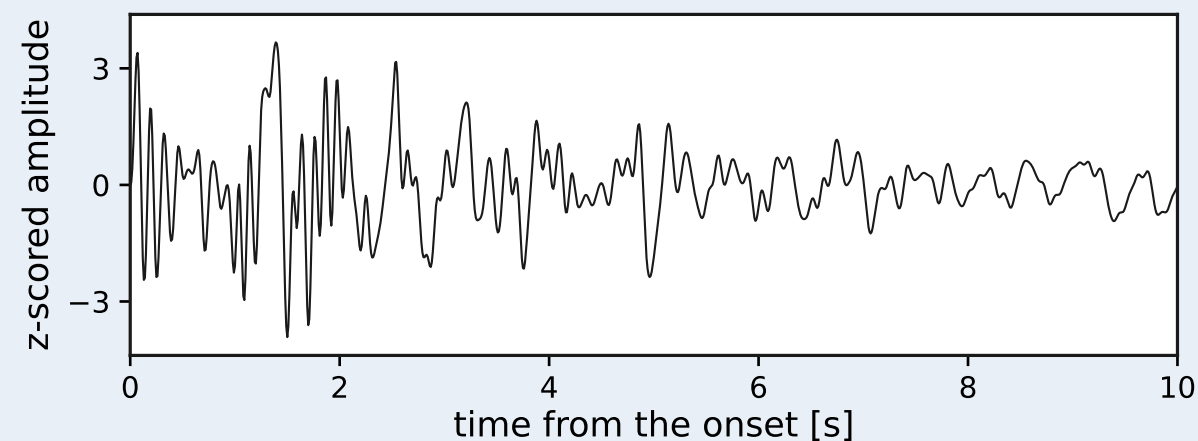
△
VN2Z



Signal conditioning

X_i

10 s from
the onset



16 descriptors

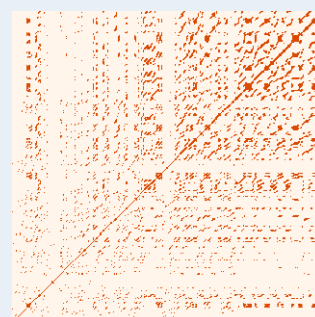
DET
 L
 L_{max}
 DIV
 L_{entr}
 LAM
 TT
 V_{max}
 V_{entr}
 W
 W_{max}
 W_{div}
 W_{entr}
 f_{peak}
 SR
 FI

RQA feature extraction

$$\vec{X}_i = [X_i, X_{i+\tau}, \dots, X_{i+(m-1)\tau}]$$

$$R_{i,j} = \Theta(\varepsilon - \|\vec{X}_i - \vec{X}_j\|)$$

13 recurrence descriptors



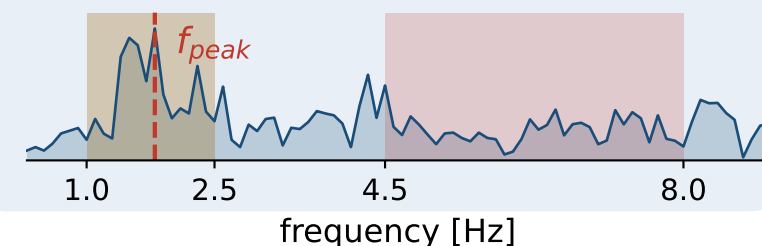
$R_{i,j}$

Spectral feature extraction

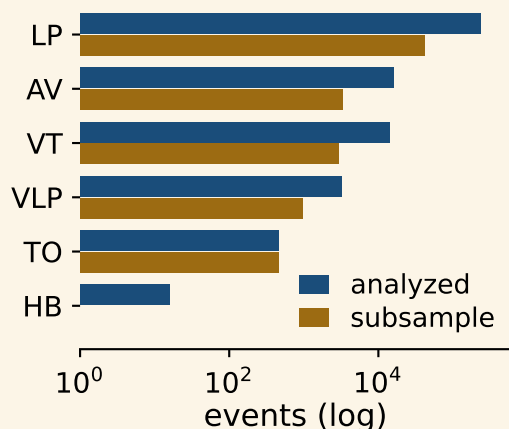
$$f_{peak} = \operatorname{argmax}_{f \in B} A(f)$$

$$SR = \bar{A}_B / \max_B A$$

$$FI = \log_{10} (\bar{A}_{high} / \bar{A}_{low})$$



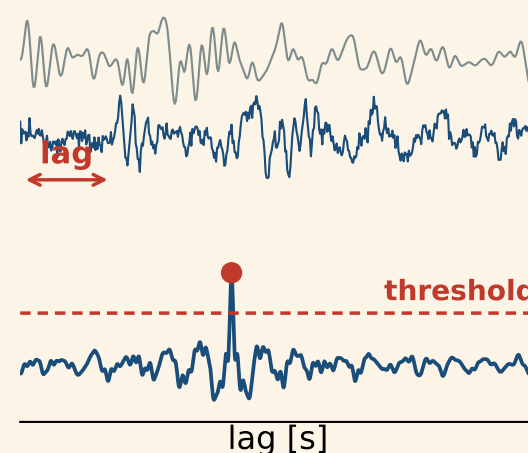
Sampling strategy



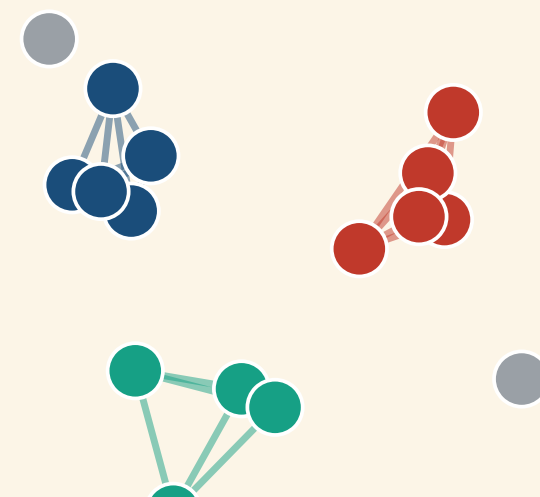
Family detection, stage 1



Family detection, stage 2



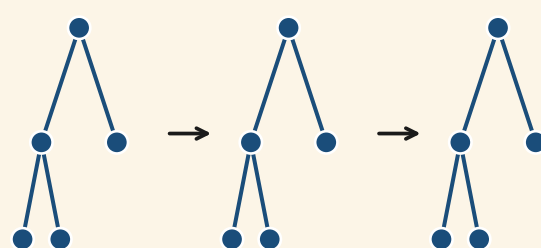
Seismic families



Data partitioning schemes



Classifier and training



$$\hat{y} = \sum_k f_k(x)$$

boosted trees, Bayesian search

Evaluation metrics

